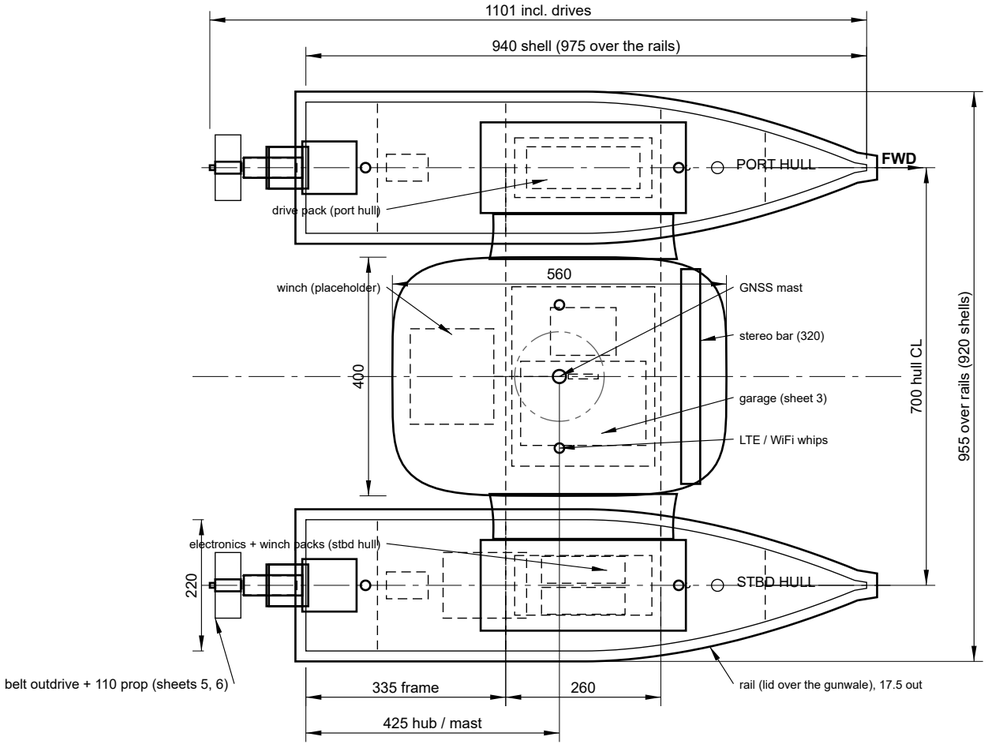
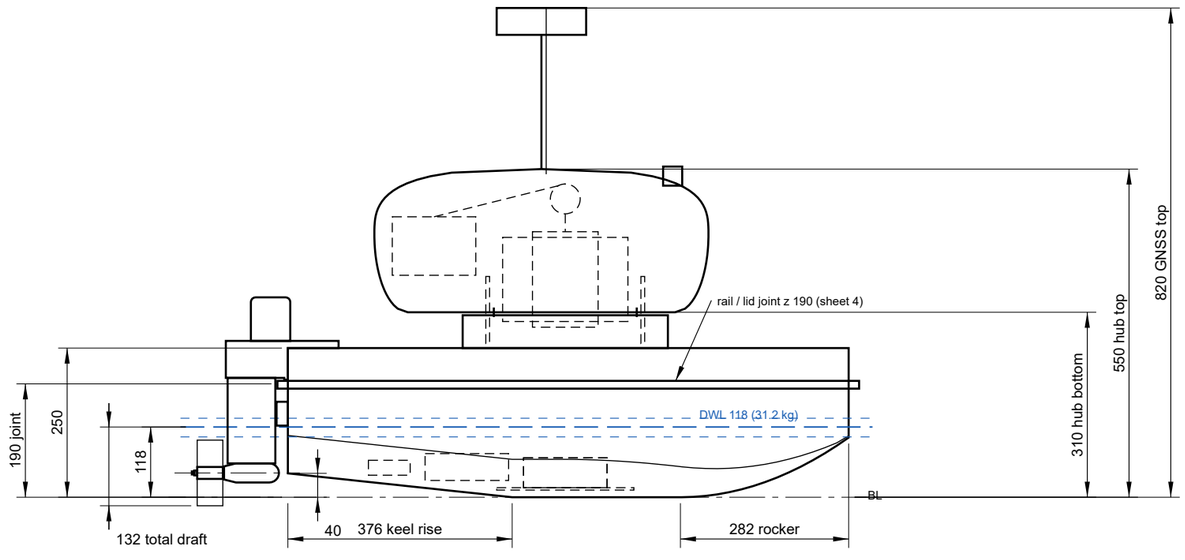


PLAN VIEW

SCALE 1:10



PROFILE (FROM STARBOARD)

SCALE 1:10

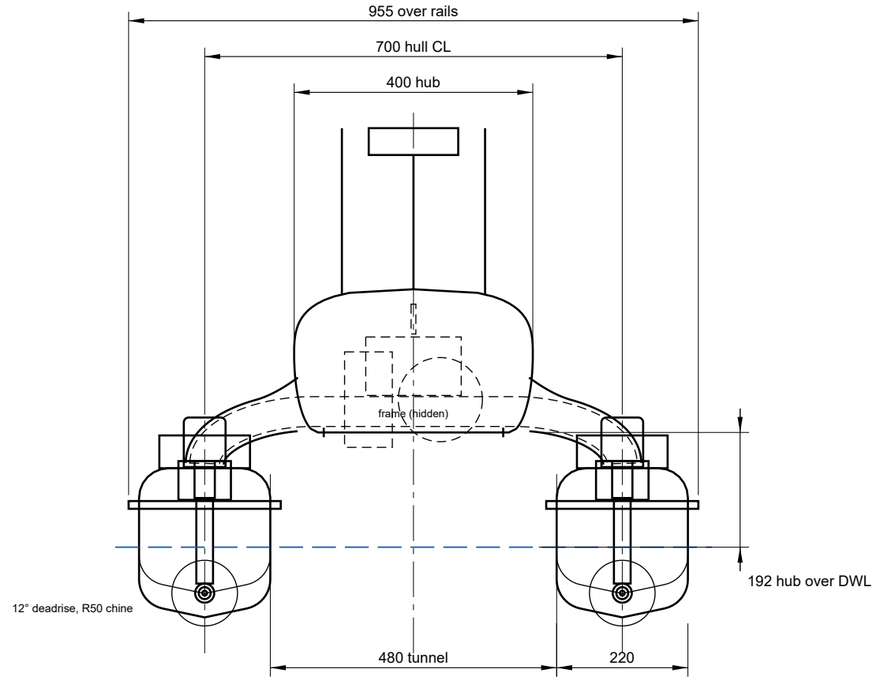
MODULES

Module	Design kg	Contents
Port tub	6.1	one-piece PLA print + ring frames, flange with 30 M4 inserts; drive pack, VESC, heat plate, leak probe, thrust pad
Stbd tub	5.6	the same shell; electronics + winch packs, Pi 5 + carrier, VESC, heat plate
Upper module	19.4	bridge (arch frames, arms, hub garage: winch, pod, Van Dorn; mast, cameras) + 2 lids + 2 belt outdrives
Boat	31.2	incl. 10% growth margin and a 2 L water sample; no robot arm

ASSEMBLY

1. Fit each tub's heat plate, packs and electronics, the leak probe (sheet 4) and the drive's thrust pad (sheet 6).
2. Bond the ribs into each lid with the foot bolted up as the jig (sheet 3). Then 8x M6 x 60 up through the rib bosses hold each foot, EPDM pad between.
3. Bolt each outdrive mount to its lid from inside: 4x M5 up through the aft deck, 2x through the aft wall.
4. Press the 3 mm cord into each tub's flange groove.
5. Hold the bridge + lids just above the tubs, plug the lid connectors inside, lower it; the drip lips locate.
6. Screw down the rails: 30 M4 per hull, 60 mm pitch within 150 mm of a frame, 100 elsewhere.

Waterlines: DWL 118 at 31.2 kg; thin blue = light 101 / max 132. Lid joint 72 above the DWL.
Wiring enters each hull only through IP68 bulkhead connectors in its lid. The packs sit low,
on aluminium plates on the hull bottoms, so the water carries their heat away.
Geometry comes from mechanical/hull_sizing.py; regenerate after changes.



VIEW FROM ASTERN

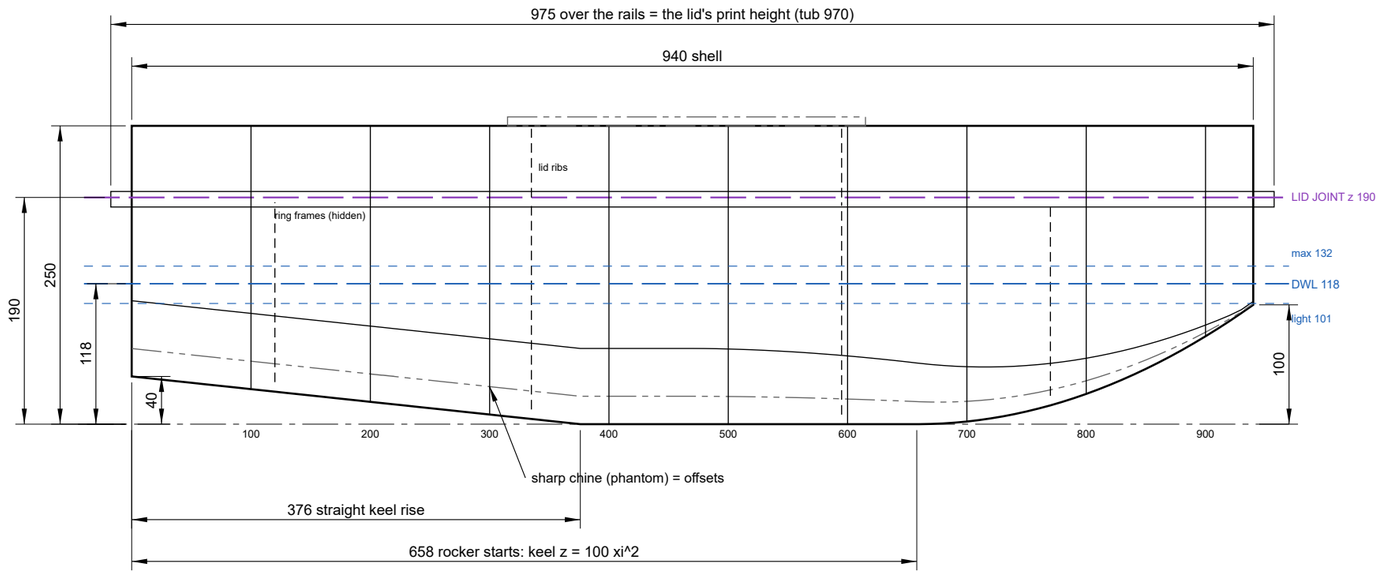
SCALE 1:10

D.R.E.A.M.S — autonomous survey catamaran

General arrangement: plan, profile, view from astern

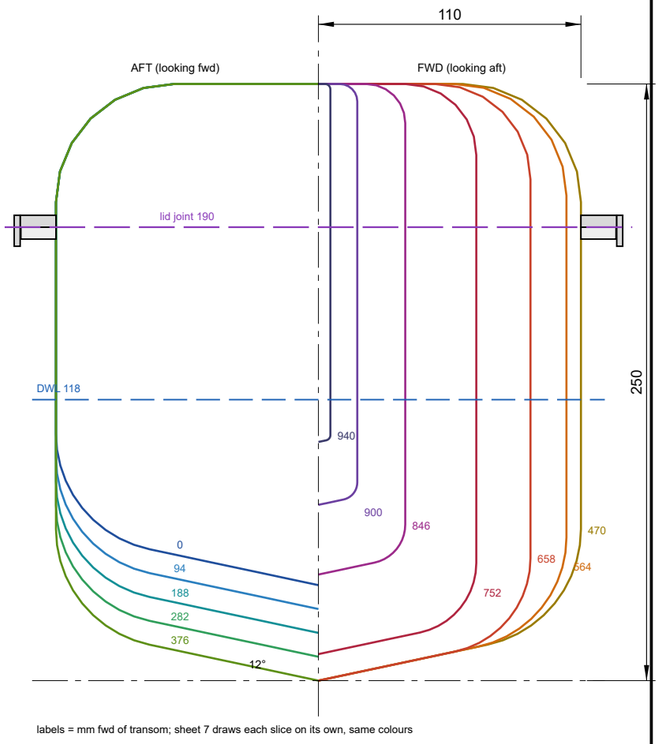
All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE 1:10	SHEET 1 OF 7	REV E (940 hulls, one-piece prints)
DATE 2026-10-05	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py



PROFILE (ONE HULL)

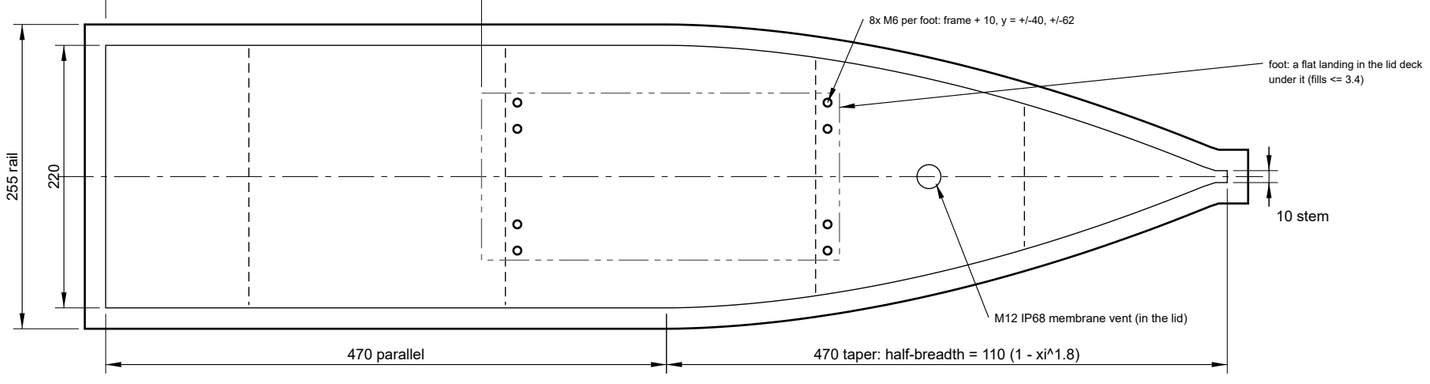
SCALE 1:5



labels = mm fwd of transom; sheet 7 draws each slice on its own, same colours

BODY PLAN (with fillets)

SCALE 1:2.5



PLAN (ONE HULL)

SCALE 1:5

OFFSETS (sharp chine; mm from transom / baseline / hull centreline)

x	keel z	chine z	half-breadth	R chine / deck
0	40.0	63.4	110.0	50 / 50
94	30.0	53.4	110.0	50 / 50
188	20.0	43.4	110.0	50 / 50
282	10.0	33.4	110.0	50 / 50
376	0.0	23.4	110.0	50 / 50
470	0.0	23.4	110.0	50 / 50
564	0.0	22.1	103.9	47 / 47

x	keel z	chine z	half-breadth	R chine / deck
658	0.0	18.9	88.9	40 / 40
752	11.1	25.2	66.1	30 / 30
800	25.4	36.4	51.8	23 / 23
846	44.4	52.2	36.4	16 / 16
900	73.6	77.1	16.3	7 / 7
940	100.0	101.1	5.0	2 / 2

CONSTRUCTION (open tub + removable lid, each printed in one piece)

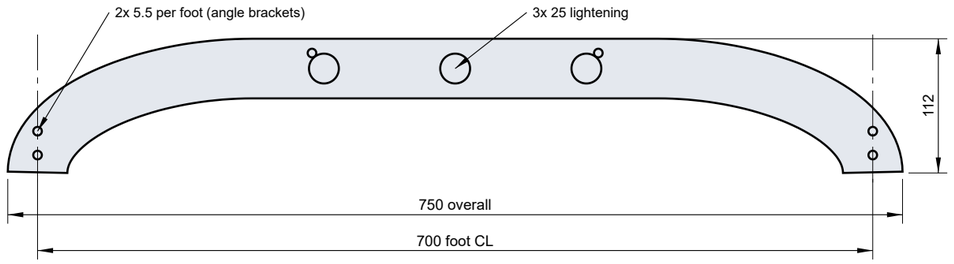
- Loft the sharp-chine hull from the offsets, fillet chine R50 and deck edge R50 (0.45 x half-breadth near the stem), split at the lid joint plane z 190.
- PLA on the Elegoo OrangeStorm Giga: 0.6-0.8 nozzle, 4 walls (2.4), +3-5 % flow. Glass outside (200 g/m2; 2nd layer on keel, chines, stem), 2 coats epoxy inside.
- Print the tub and the lid standing on the stern: 970 and 975 tall incl. the rail (Giga: 1000). The rail laps 15 / 17.5 past the transom, so support the 4 mm transom (lid: aft wall) off the bed up to the flange. 10 mm brim; slow the walls near the top.
- Tub: ring frames at x 120 / 335 / 595 / 770, 8 deep, aft face 45 deg so they print unsupported, a limber notch at the keel (sheet 3).
- Lid: 8 mm ribs at x 335 / x 595 with the foot-bolt bosses, printed flat and bonded in with thickened-epoxy fillets (sheet 3).
- Rail: 15 mm flanges, 3 mm cord in a 5 x 2.2 groove, 30x M4 into heat-set inserts (sheet 4); epoxy and sand the flange: layer lines cross the gasket.
- One sealed space per hull: leak probe at the keel low point (carrier J8 port / J9 stbd), M12 vent in the lid. Epoxy only, slow hardener: PLA softens ~55 C.
- Shell 3.4 kg per hull incl. ribs, ring frames, glass, epoxy, + 0.8 kg rail. Pressure-test (<= 0.03 bar) before fitting out.

D.R.E.A.M.S — autonomous survey catamaran

Demi-hull lines, lid joint plane, one-piece prints, offsets and fillets

All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

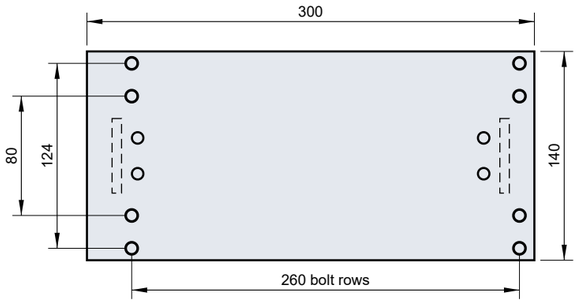
SCALE 1:5 / 1:2.5	SHEET 2 OF 7	REV E (940 hulls, one-piece prints)
DATE 2026-10-05	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py



1/4 in (6.35) 6061-T6, 2 req. Section 50 deep, +/-25 from the arch centreline; feet stand on the foot plate, 7.5 above the lid deck. Cut file: cad/arch_frame_plate.dxf

ARCH FRAME PLATE (FROM AHEAD)

SCALE 1:5



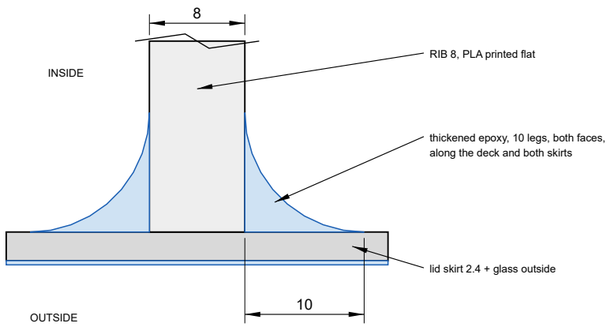
6 mm 6061. Frames (hidden) at -130 / +130. 8x M6 tapped through at frame + 10
(bolts come up from inside the lid); 2x 5.5 per angle bracket. It sits on a flat landing in the lid deck

FOOT PLATE (PLAN, 2 req)

SCALE 1:4

Arch centreline (stbd arm)	y	z
hub span, straight	0 to 170	345
Bezier P0 (arm root)	170	345
Bezier P1	260	345
Bezier P2	350	308
Bezier P3 (foot)	350	257.5

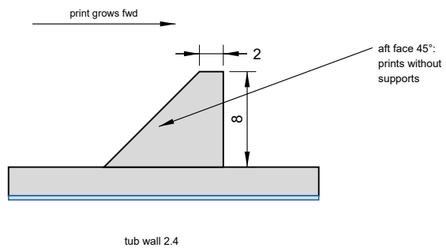
Port arm is the mirror image.
Tangent horizontal at P0, vertical at P3.



RIB BOND (LID SKIRT CUT AT MID-DEPTH, FROM ABOVE)

SCALE 2:1

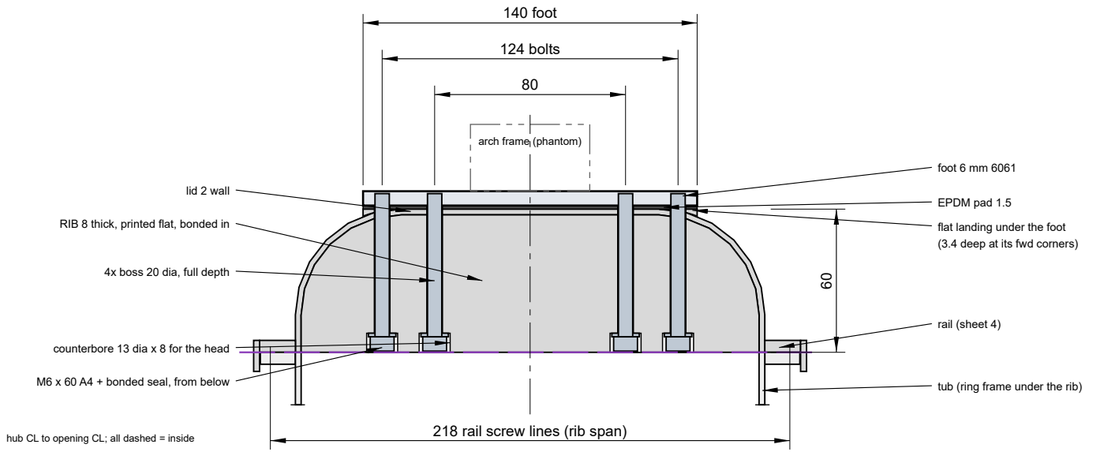
Fit each rib with the foot bolted up through the deck and the rib bosses: the foot is the jig, so the holes line up. Sand the PLA, fillet both faces along the deck and both skirts, then coat the inside.



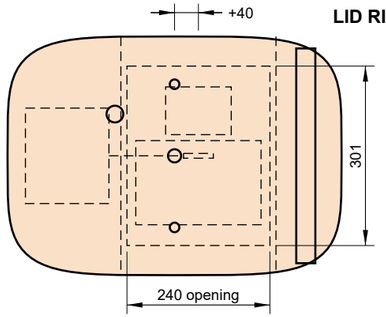
TUB RING FRAME (SECTION ALONG THE HULL)

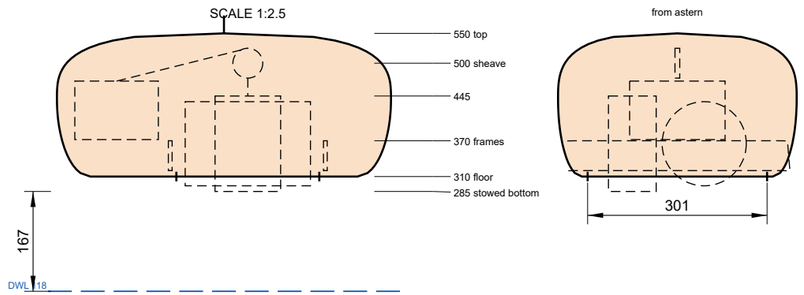
SCALE 2:1

At x 120 / 335 / 595 / 770, all the way round inside the tub,
up to the joint plane. 12 x 6 limber notch at the keel.



LID RIB (SECTION AT x 605, LOOKING FWD; THE AFT RIB IS THE SAME)





HUB GARAGE: WINCH, POD, VAN DORN (PLAN, SIDE, FROM ASTERN)

SCALE 1:10

Pod 110 x 80 x 160 (port) and a 2 L horizontal Van Dorn, 5 in PVC, 141 x 210 (stbd), side by side under the sheave: margins +30 / +70 mm through the opening. Winch, pod and sampler are placeholders until they're sized; the winch sits on a bracket off the aft frame, the sheave over the opening on the garage holes in the frames.

HARDWARE (whole boat)

Qty	Item
2	Arch frame plate, 1/4 in 6061-T6, laser / waterjet cut (DXF)
2	Foot plate, 6 mm 6061, 300 x 140, 8x M6 tapped
4	Angle bracket 25 x 25 x 3 Al, 40 long (frame to foot)
16	M6 x 60 A4 socket cap + M6 bonded seal washer (feet, from inside)
2	EPDM pad 1.5 mm, 300 x 140 (under each foot)
60	M4 x 12 A4 pan head + washer, and M4 x 8.1 brass heat-set insert (rails)
2	3 mm EPDM / silicone cord, 2.14 m (one loop per tub)
1	Hub cowl, 2 arm fairings, 2 foot shoes (printed PLA, painted light)
4	Lid rib, 8 mm PLA printed flat, 100 % round the bosses; 2 bonded into each lid
1	Garage: winch bracket off the aft frame, sheave bracket, printed docking cone
2	Belt outdrive, complete (sheets 5, 6)
2	M12 IP68 membrane vent (one per lid)
4	IP68 bulkhead connector in the lids: motor (x 100) and bridge (x 625), per hull
2	Leak probe at each tub's keel low point (carrier J8 port / J9 stbd)

LOAD PATH

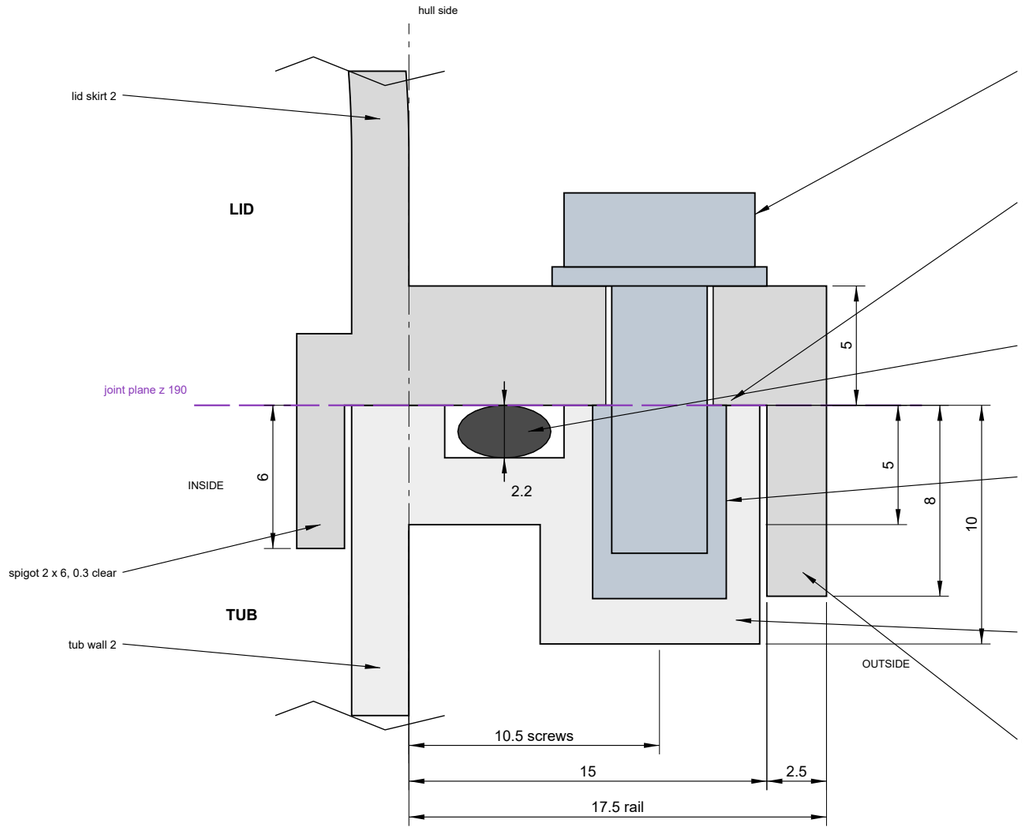
Each frame is a portal fixed at its feet.
Worst case: lifted by one hull x1.5
100 N m per frame -> 38 MPa,
SF 7.3 on 6061-T6 yield.
Foot: pivots on its far edge; the 8
M6 in the rib bosses hold it: SF 4.9.
Lid rib carries it to the rail: SF 6.6;
its bonded ends: SF 11.7.
Rail: 3 M4 inserts per frame: SF 6.2.
Nothing goes through a tub: all
hardware bolts to the lids.
Details: docs/hull_and_crossbeam_sizing.md

D.R.E.A.M.S — autonomous survey catamaran

Arch frame, foot plate, lid rib, hub garage, rib bond, ring frame

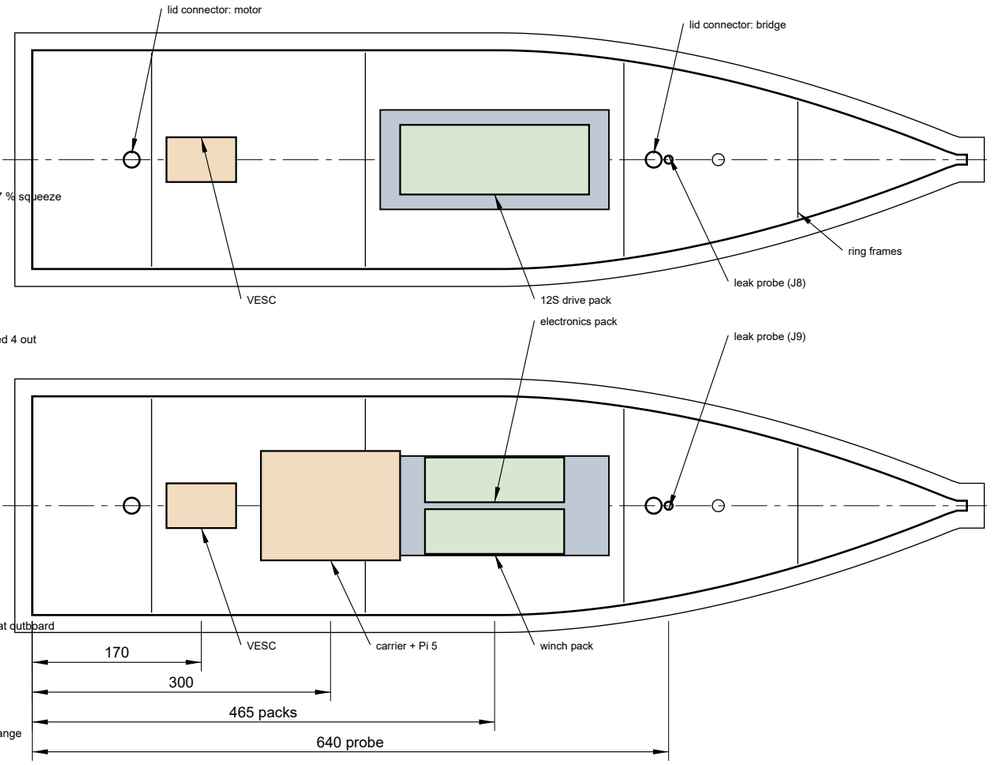
All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE as noted	SHEET 3 OF 7	REV E (940 hulls, one-piece prints)
DATE 2026-10-05	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py



RAIL: LID OVER THE GUNWALE (SECTION THROUGH A SCREW)

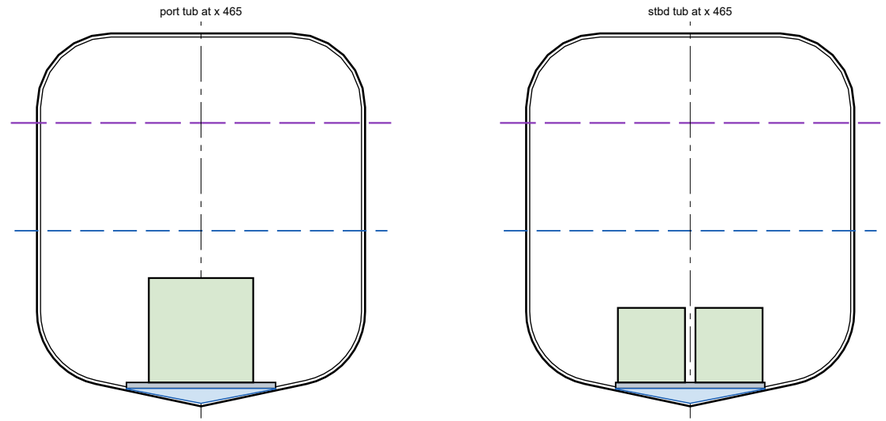
SCALE 4:1



INSIDE THE TUBS (PLAN, PROPOSED)

SCALE 1:6

SECTIONS AT x 465, looking fwd, 1:4: the heat plate is bedded on thickened epoxy (blue) on the V bottom, the packs sit on the plate. DWL dashed blue, lid joint purple.



WIRING (everything enters a hull through its lid connectors)

- 12S: drive pack (port) -> port VESC; to the stbd VESC across the bridge (port arm, hub, stbd arm).
- CAN: carrier (stbd) -> stbd VESC -> across the bridge -> port VESC (node IDs 1, 2).
- Electronics pack (stbd) -> carrier -> Pi 5 on the carrier's 5 V. Never also power the Pi from USB-C.
- Winch pack (stbd) -> carrier's fail-safe winch kill -> up the stbd arm -> winch in the hub garage.
- Pod tether (RS-485 + pack voltage) from carrier J2 -> up the stbd arm -> the winch's slip ring.
- Mast (GNSS, LTE, WiFi), stereo cameras, e-stop -> down the stbd arm. Leak probes: J8 port, J9 stbd.
- Motor wires: outdrive -> lid connector at x 100 -> that hull's VESC.

HEAT AND WATER

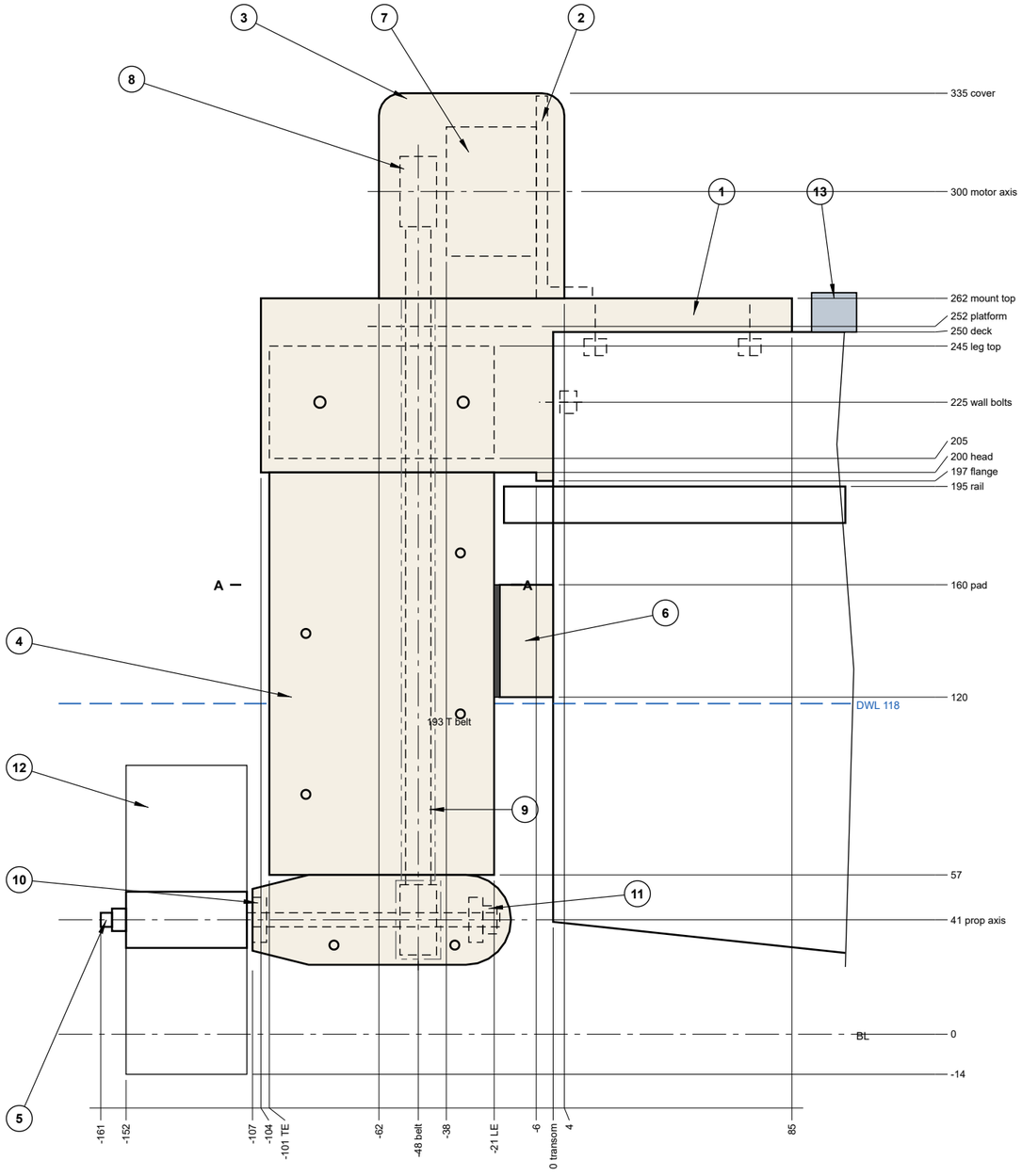
- VESCs and packs sit on 6061 plates bedded on the hull bottom, so the water takes the heat; PLA softens ~55 C. Log the hull air temperature (the carrier's SHT45s).
- Limber notches in the ring frames let a leak run along the keel to the probe at x 640, fwd of the packs.
- Lifting a lid: unscrew the rail, raise it ~100 mm on the connector slack, unplug inside, lift away.

D.R.E.A.M.S — autonomous survey catamaran

Hull lid joint (rail) and what goes inside each hull

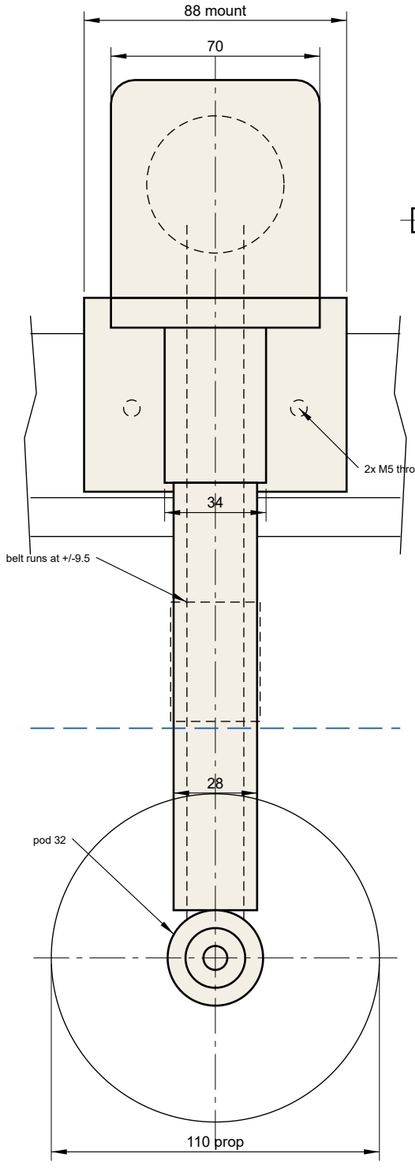
All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE as noted	SHEET 4 OF 7	REV E (940 hulls, one-piece prints)
DATE 2026-10-05	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py



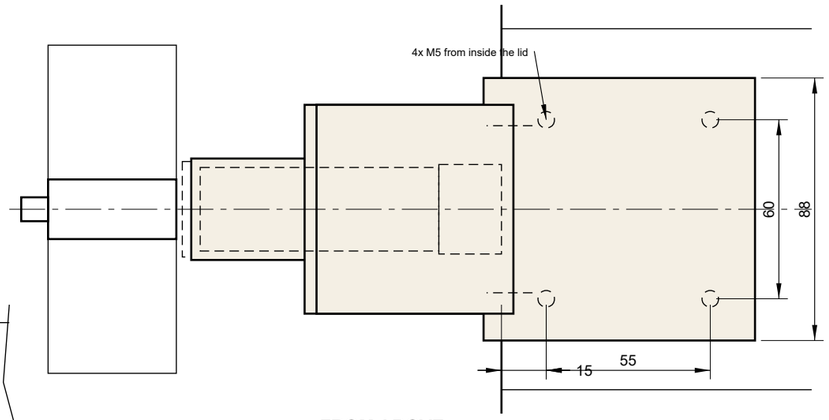
SIDE VIEW, STARBOARD DRIVE (FIXED)

SCALE 1:2



FROM ASTERN

SCALE 1:2



FROM ABOVE

SCALE 1:2

KEY POINTS (mm; x fwd of the transom, z above the baseline, y off the hull CL)

Feature	x	z	y
Mount: deck flange	0 to 85	250 to 262	+/-44
Mount: flange down the lid's aft wall	-6 to 0	197 to 262	+/-44
Mount: head block	-104 to -6	200 to 262	+/-17
Mount: platform (under the cover)	-66 to -6	252 to 262	+/-35
Mount: leg socket	-101 to -21	205 to 245	+/-14
Deck bolts, 4x M5 up from inside	15, 70	250	+/-30
Wall bolts, 2x M5 fwd from inside	10 to -6	225	+/-28
Thrust pad (+ 2 EPDM face)	-21 to 0	120 to 160	+/-15
Motor plate, 4 mm 6061	-6 to -2	262 to 334	+/-25
Motor can (41xx)	-38 to -6	axis 300, dia 46	0
Motor cover	-62 to 4	262 to 335	+/-35
Belt plane (both pulleys)	-48	300 / 41	runs +/-9.5
Leg, NACA 0035, 80 x 28	TE -101, LE -21	57 to 245	+/-14
Lower pod, dia 32	tail -107, nose -15	axis 41	0
Bearings 625-2RS	-30 to -25; -107 to -102	axis 41	
Shaft collar / shaft, dia 5	-25 to -20 / -161 to -19	axis 41	
Prop hub 20 / disc 110 / nylock	-152 to -109 / -157	41 +/- 55	
Motor-wire connector (lid deck)	100	250	0

PARTS (the balloons; 1-6 are drawn on sheet 6, 7-13 are bought)

1. Mount
2. Motor plate
3. Motor cover
4. Leg + pod halves
5. Prop shaft
6. Thrust pad
7. Motor
8. Pulleys x2
9. Belt
10. Bearings x2
11. Shaft collar
12. Prop, washer, nylock
13. Lid connector

HOW TO USE THIS SHEET

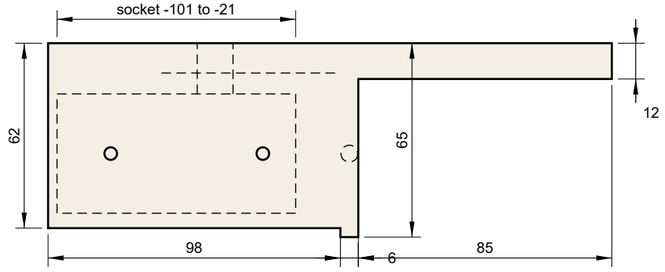
- Datums: x from the transom (the lid's aft face), z from the baseline, y from the hull centreline. Model the drive in place in the boat assembly.
- Fixed: no pivot. The leg runs 3.5 mm clear of the rail (as close as the rail allows); its nose rests on the thrust pad on the transom. Tightest gap: wall flange over rail 2.0 mm.
- At full thrust (55 N) the pad takes 119 N and the mount 64 N; the leg bends 5.5 N m at the pad instead of 10.1 N m at the mount. Reverse hangs on the bolts.
- Belt tension: slide the motor up its slotted plate (+/-4). The port drive is the mirror image with an LH prop. Parts, sizes and the motor note: sheet 6.

D.R.E.A.M.S — autonomous survey catamaran

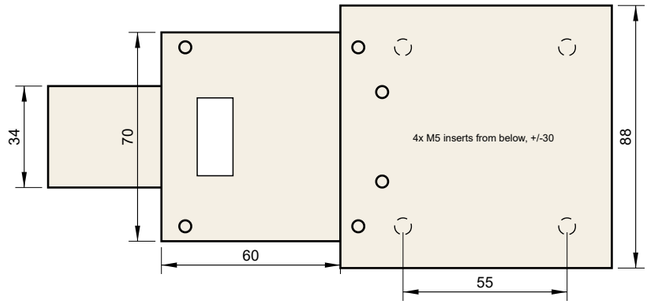
Belt outdrive (fixed), assembled: side, from astern, from above, key points (for CAD)

All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE 1:2	SHEET 5 OF 7	REV E (940 hulls, one-piece prints)
DATE 2026-10-05	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py

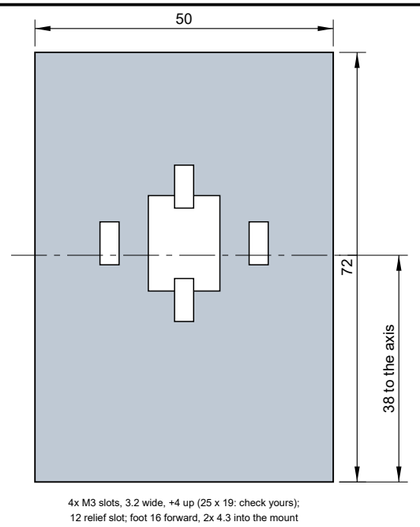


socket = leg section + 0.2, 40 deep, 2x M4 across; wall flange: 2x M5 inserts



1 MOUNT (side, above)

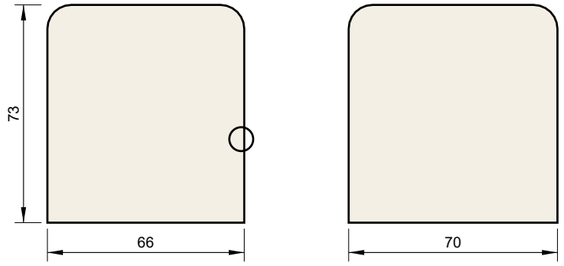
SCALE 1:2



4x M3 slots, 3.2 wide, +4 up (25 x 19: check yours);
12 relief slot; foot 16 forward, 2x 4.3 into the mount

2 MOTOR PLATE (4 mm 6061, L)

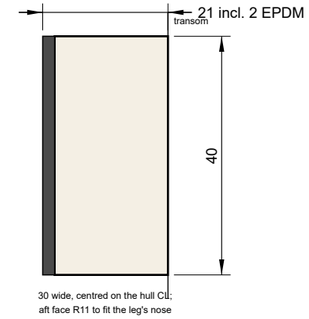
SCALE 1:1



2 wall, R8 edges, open at the bottom; 4x M3 tabs into the mount top; 8 grommet in the front wall for the motor wires; 10 x 3 drain slot at the front.

3 MOTOR COVER (side, astern)

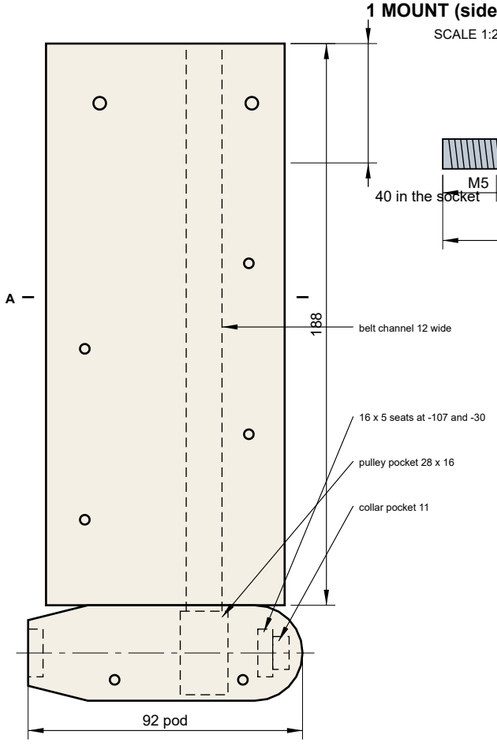
SCALE 1:2



30 wide, centred on the hull CL;
aft face R11 to fit the leg's nose

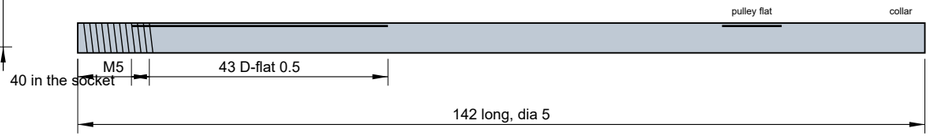
6 THRUST PAD

SCALE 1:1



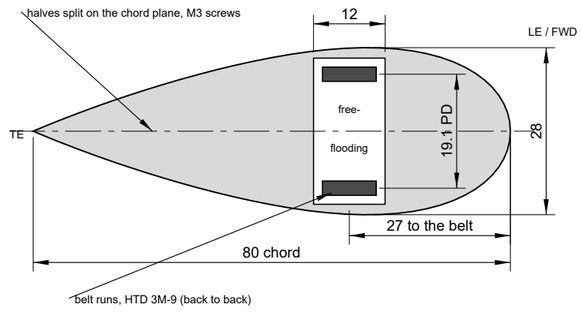
4 LEG + POD HALF (stbd half; split on y = 0)

SCALE 1:2



5 PROP SHAFT (316, ends at x -161 and -19)

SCALE 1:1



SECTION A-A THROUGH THE LEG

SCALE 1:1

PURCHASED PARTS (per drive; 2 drives)

No.	Qty	Item
7	1	Outrunner, 41xx (46 x 32 can), 25 A+. KV: see the note
8	2	HTD 3M pulley, 20T (19.1 PD), 9 mm belt; bores to the motor / 5 mm shaft
9	1	HTD 3M belt, 9 mm, 193 T (579 mm), PU with aramid or stainless cords
10	2	625-2RS bearing, stainless (5 x 16 x 5): they run wet
11	1	Shaft collar, 5 mm bore, 10 OD x 5, set screw
12	1	Prop 110 (props/toroidal_..._RH_shaft stbd, _LH_ port) + washer + M5 nylock A4
13	1	IP68 bulkhead connector, 3-pin, in the lid's aft deck at x 100
-		6x M5 x 20 A4 + washers (4 up through the deck, 2 through the aft wall), 3 mm Al backing plates
-		6x M5 heat-set inserts in the mount; 2x M4 x 40 A4 + nylocks across the socket
-		~12x M3 A4 + inserts (leg/pod halves, cover); 2x M4 x 8 (motor plate foot)

NOTES

- Print 1, 3 and 4 in ABS or ASA (the EXP S5 lists ABS), not PLA: the motor and the sun heat them. 6 walls or 100 % infill round the socket and the inserts. The mount can be milled from 6061 later.
- Bond 6 to the transom with thickened epoxy; it carries the forward thrust. The leg lifts off with the lid; nothing on the drive bolts to the tub.
- Loads at 55 N forward: pad 119 N, mount 64 N, leg 5.5 N m at the pad (10.1 N m with no pad).
- MOTOR KV. V3's drive rail is 12S (~44 V). Rev C picked 380 KV for an 18 V pack; on 12S it would run at ~10-25 % duty from cruise to top speed. ~150-200 KV suits 12S. Most motors with enough power at that KV are a size up (50xx, ~55 dia): that fits the 70 cover with the motor axis ~5 mm up. The drawings keep the 41xx envelope until you pick one.

MODELING ORDER (SolidWorks)

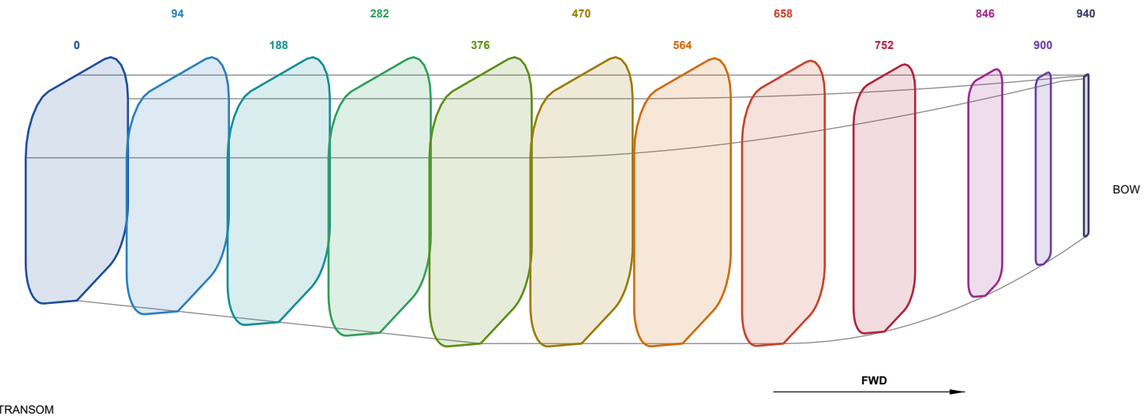
1. Drive assembly layout sketch on the Front plane: transom (x 0), deck (z 250), belt plane, motor and prop axes. Origin = transom / baseline / hull CL, like the hull.
2. Mount: extrude the side profile +/-17 (head block); add the deck and wall flanges +/-44 and the platform +/-35; cut the socket (leg section + 0.2), the belt slot and the holes.
3. Leg + pod: section A-A at z 57 extruded to 245; pod revolved on the prop axis; combine, cut the pockets, split on y = 0. Mirror the half for the other side.
4. Pad, plate, cover; then mate everything fixed and run Interference Detection.

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Belt outdrive (fixed), part by part (for CAD), purchased parts and modeling order

All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE as noted	SHEET 6 OF 7	REV E (940 hulls, one-piece prints)
DATE 2026-10-05	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py



WHERE EACH SLICE IS (ONE HULL)

SCALE sketch, not to scale

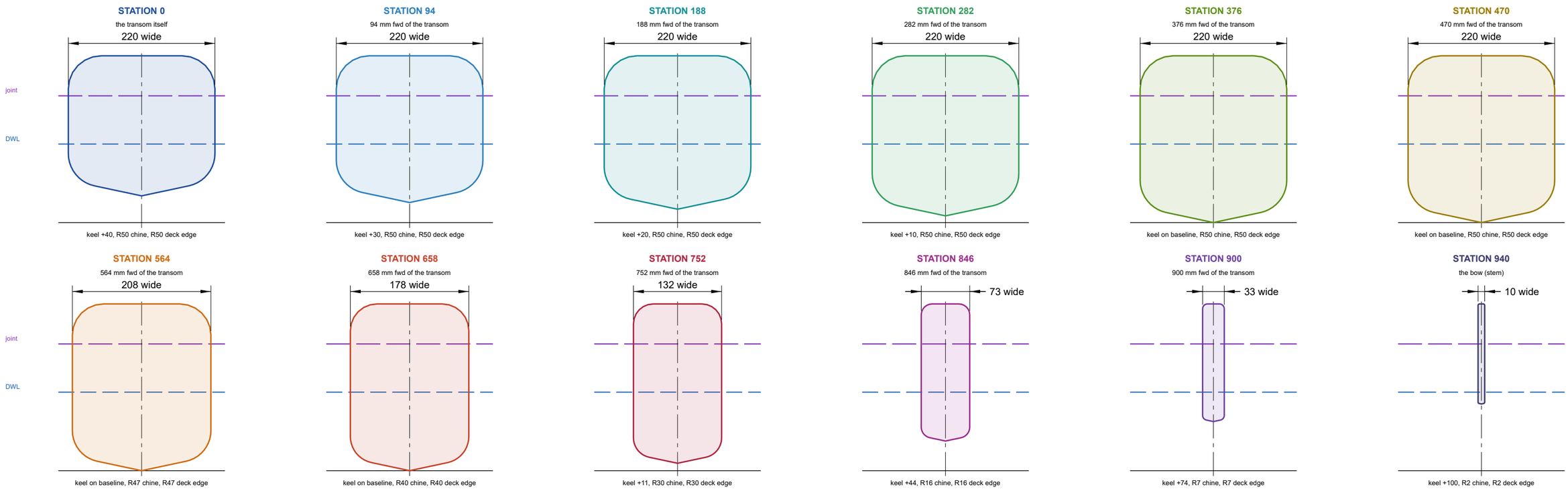
HOW TO READ THE BODY PLAN (SHEET 2)

Each coloured shape is one slice straight across ONE hull, like a slice of bread. Its number is where the slice is: mm forward of the transom.

Sheet 2 stacks all the slices on one drawing. The hull is the same on both sides, so it draws only half of each slice: the stern half of the boat on the left of the centreline, the bow half on the right, in these colours.

Reading from stern to bow:
- 0 to 376: the same shape, but the keel lifts toward the transom (+40 mm at 0), so the stern draws less water.
- 376 to 470: the widest part (220 mm), keel on the baseline.
- 470 to 940: it narrows toward the bow; from 658 the keel also rises, to +100 at the bow, which ends in a 10 mm flat face.

In each box: blue dashes = waterline (118), purple = lid joint (190).
In SolidWorks, each box is one sketch on a plane at that x (guide, A2).



Scale 1:6. Full slices, both halves, looking forward from the stern.

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Hull sections one at a time: a reading aid for the body plan

All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE 1:6 / sketch	SHEET 7 OF 7	REV E (940 hulls, one-piece prints)
DATE 2026-10-05	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py